LESSON 3.5

Compare Ratio Relationships

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can compare ratios by finding unit rates or using equivalent ratios.

WHY THIS MATTERS Two friends used the paint shown to create shades of purple paint. Whose mixture contains the most blue paint?

REMEMBER

Compare Ratio Relationships

Comparing Ratio Relationships To compare ratios fairly, make one amount the same in both, or find the unit rate (the amount for just 1). Then you can see which ratio is bigger.

1. Convert each ratio to a unit rate (e.g. price per item or miles per gallon).
2. Alternatively, scale both ratios to a common term (like a common denominator).
3. Compare the resulting unit rates to identify the better buy or faster rate

Words: Unit rate *(Tasa unitaria)* · Equivalent ratio *(Razón equivalente)* · Compare *(Comparar)* · Simplify *(Simplificar)* · Common denominator *(Denominador común)*

1 **Circle the letter of the best answer. Show how you know.**

A ratio table for 3:2 gives the points (3, 2) and (6, 4). Which point also lies on the same graph?

(A) (2, 3)
(B) (8, 6)
(C) (9, 6)
(D) (9, 5)

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

The graph of a set of equivalent ratios always passes through which point?

(A) (0, 0)
(B) (0, 1)
(C) (1, 1)
(D) (1, 0)

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

On a graph of cups of juice (x) to cups of water (y), the point (4, 10) is plotted. What ratio does that point show?

(A) 4:10
(B) 4:14
(C) 10:4
(D) 14:10

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A ratio graph passes through (2, 5). Which point is NOT on the same line?

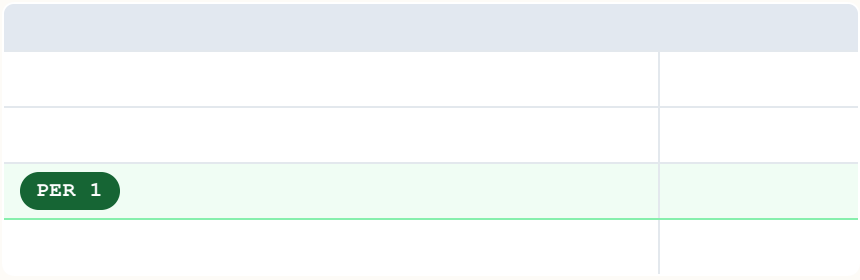
(A) (4, 10)
(B) (6, 15)
(C) (8, 20)
(D) (10, 20)

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Two pizza shops are running deals. Mario's Pizza sells 2 large pizzas for $18. Sal's Pizza sells 3 large pizzas for $25. A school needs to buy pizza for a class party and wants to get the best deal per pizza. What is Mario's price per pizza?

(A) $4.50
(B) $8.33
(C) $9.00
(D) $18.00



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

6 **Circle the letter of the best answer. Show how you know.**

Two pizza shops are running deals. Mario's Pizza sells 2 large pizzas for $18. Sal's Pizza sells 3 large pizzas for $25. A school needs to buy pizza for a class party and wants to get the best deal per pizza. What is Sal's price per pizza?

(A) about $7.50
(B) about $8.33
(C) about $9.00
(D) about $12.50

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The two recipes use different amounts, so you cannot compare them directly. How did building equivalent ratios let you compare them fairly?

SENTENCE STARTER I scaled both ratios to the same amount of ___, then compared the ___; Chef ___'s recipe is more chocolatey because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each Chef Academy ratio comparison to the correct winner.

_____ Smoothie A: 2 cups fruit for 6 cups water vs. Smoothie B: 3 cups fruit for 8 cups water — more fruity?

_____ Bagels: 4 for $5 vs. 6 for $9 — better price per bagel?

_____ Trail mix: 6 oz nuts to 10 oz oats vs. 9 oz nuts to 15 oz oats — more nutty?

_____ Dressing: 3 tbsp oil to 4 tbsp vinegar vs. 5 tbsp oil to 8 tbsp vinegar — more oily?

(A) Smoothie B (0.375 > 0.333 fruit per cup)

(B) 4 for $5 ($1.25 < $1.50 per bagel)

(C) Equal (both simplify to 3:5)

(D) 3:4 (equals 6:8, greater than 5:8)

2 **Write the name of the correct group on each line.**

Each chef compared two recipes. Sort their conclusions into Correct or Incorrect.

Groups: Correct Comparison | Incorrect Comparison

_____ Sauce 5:6 vs. 7:9 — chef says 5:6 is thicker (more tomato)

_____ Punch 2 cups juice for 5 cups water vs. 3 for 8 — chef says the 2:5 punch is more juicy

_____ Frosting 3:4 sugar-to-butter vs. 5:8 — chef says 3:4 is sweeter

_____ Cocoa 3:7 vs. 2:5 — chef says 3:7 is weaker (less cocoa per oz)

_____ Tea 4:5 leaves-to-water vs. 5:7 — chef says 5:7 is stronger

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Two pizza shops are running deals. Mario's Pizza sells 2 large pizzas for $18. Sal's Pizza sells 3 large pizzas for $25. A school needs to buy pizza for a class party and wants to get the best deal per pizza. Which shop is the better deal?

(A) Mario's, because $18 is less than $25

(B) Mario's, because 2 is less than 3

(C) Sal's, because $8.33 per pizza is less than $9.00

(D) They cost the same

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Chef Academy's snack bar sells 3 mini quiches for $6 or 5 mini quiches for $8. Which is the better deal per quiche?

(A) 3 quiches for $6

(B) 5 quiches for $8

(C) They cost the same

(D) Cannot determine

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Two soup recipes use salt. Recipe A uses 4 tsp salt for 10 cups of broth. Recipe B uses 3 tsp salt for 8 cups of broth. Which soup is saltier per cup?

(A) Recipe B

(B) They are equally salty

(C) Cannot determine

(D) Recipe A

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Chef Mara's lemon glaze uses 6 oz of lemon juice for 9 cupcakes. Chef Devi's uses 4 oz for 7 cupcakes. Whose glaze has more lemon per cupcake?

(A) Chef Devi
(B) Chef Mara
(C) Same amount
(D) Cannot determine

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The two recipes use different amounts, so you cannot compare them directly. How did building equivalent ratios let you compare them fairly?

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